HYPERELLIPTICITY FROM SEVERAL BORCHERDS DIVISORS, BY COUNTING TORSION

1. The problem this removes

Guo and Yang's method for the equation of a Shimura curve quotient produces functions as Borcherds lifts: a weakly holomorphic form f of weight $1/2$ with prescribed principal part lifts to a meromorphic modular form Ψ_f whose divisor is the Heegner divisor $Z(f)$ prescribed with it. What the lift does not give for free is that Ψ_f is a *function* on the quotient. Their Assumption 40 asks exactly this, and it is a genuine hypothesis: the character of Ψ_f has finite order, so a power of Ψ_f is a function and the class of $Z(f)$ is torsion in the Jacobian, but the class need not be trivial.

So a search that returns one divisor $Z(f)$ of degree two proves hyperellipticity only under the assumption. The observation below is that a search returning *several* divisors proves it outright, as soon as there are more of them than the rational torsion of the Jacobian can separate. The hypothesis is then replaced by a count, and both sides of the count are computable without a model of the curve.

2. The criterion

Throughout, X is a smooth projective geometrically connected curve over $\mathbb{Q}$ of genus $g \geq 2$ with $X(\mathbb{Q}) \neq \emptyset$, and $J = \mathrm{Jac}(X)$. Since X has a rational point, the natural map $\mathrm{Pic}^0(X)(\mathbb{Q}) \to J(\mathbb{Q})$ is an isomorphism, so a $\mathbb{Q}$-rational divisor class of degree zero is a point of $J(\mathbb{Q})$.

Lemma 1 (Borcherds)**.** *Let f be a weakly holomorphic form whose principal part is integral, and let Ψ_f be its Borcherds lift on X, with divisor $Z(f)$. Then the class of $Z(f)$ is torsion in $J(\mathbb{Q})$.*

Proof. Integrality of the principal part is the hypothesis under which the character of Ψ_f has finite order, say n; see [1, proof of Theorem 4.1], with the correction [2] restoring the finite-order condition that the printed statement omits from the definition of a principal Heegner divisor. Then Ψ_f^n has trivial character and is a rational function on X, so $n \cdot Z(f) = \mathrm{div}(\Psi_f^n)$ is principal. The divisor $Z(f)$ is a sum of Heegner cycles and so is Galois-stable, hence defines a class in $\mathrm{Pic}^0(X)(\mathbb{Q}) = J(\mathbb{Q})$, which is therefore killed by n. □

Corollary 2. *Let f have* rational *principal part, with n a positive integer clearing its denominators, and suppose the Borcherds lift of nf has divisor nZ. Then the class of Z is torsion in $J(\mathbb{Q})$.*

Proof. nf has integral principal part, so by Lemma 1 the class of nZ is torsion. A class whose n-th multiple is torsion is itself torsion, $J(\mathbb{Q})$ being finitely generated. □

Theorem 3. *Let t be an integer with $\#J(\mathbb{Q})_{\mathrm{tors}} \leq t$. Suppose $D_1, \ldots, D_r$ are pairwise distinct effective divisors of degree two on X, each defined over $\mathbb{Q}$, such that for all $i \neq j$ some positive integer multiple of $D_i - D_j$ is the divisor of a Borcherds form with integral principal part. If $r > t$, then X is hyperelliptic.*

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Proof. By Lemma 1 and Corollary 2 the classes $[D_i - D_1]$, for $i = 1, \dots, r$, all lie in $J(\mathbb{Q})_{\mathrm{tors}}$, a set of at most t elements. As $r > t$, two of them coincide: there are $i \neq j$ with $[D_i - D_1] = [D_j - D_1]$, hence $[D_i - D_j] = 0$ and $D_i \sim D_j$.

Since $D_i \neq D_j$, the linear equivalence is realised by a nonconstant $h \in \mathbb{Q}(X)$ with $\mathrm{div}(h) = D_i - D_j$. Write G for the greatest common divisor of the effective divisors D_i and D_j, so that the polar divisor of h is $D_j - G$, of degree $2 - \deg G$ with $\deg G \in \{0, 1\}$; it is not 2, as $D_i \neq D_j$. The degree of h as a morphism $X \to \mathbb{P}^1$ is the degree of its polar divisor. If $\deg G = 1$ that degree is one and $X \cong \mathbb{P}^1$, contradicting $g \geq 2$. Hence $\deg G = 0$ and h has degree two, so X is hyperelliptic. □

Remark 4. Only $r > t$ is used, not which pair of divisors becomes equivalent, and the proof does not identify the pair. For the *equation* of X one must know it, and there the candidates are few and can be separated by a different means; see Remark 8.

Remark 5. The argument is not special to degree two. With $D_1, \dots, D_r$ effective of degree d and $r > t$, the same count gives $D_i \sim D_j$ and hence a map $X \to \mathbb{P}^1$ of degree $d - \deg G \leq d$, so the gonality of X is at most d. Degree two is the case where the conclusion is a dichotomy, because a map of degree one is excluded by the genus.

3. Why both sides of the count are computable

The criterion is useful here because neither hypothesis needs a model of X.

The torsion bound. For an abelian variety over $\mathbb{Q}$ with good reduction at an odd prime p, reduction is injective on the rational torsion [4, Appendix]. Hence $\#J(\mathbb{Q})_{\mathrm{tors}}$ divides $\#J(\mathbb{F}_p) = P_p(1)$, where P_p is the numerator of the zeta function of X over $\mathbb{F}_p$, and so divides the greatest common divisor of these over any set of such primes. For a quotient of a Shimura curve the point counts over $\mathbb{F}_{p^k}$ come from the trace formula, so P_p and the bound are computed with no equation for X in sight.

The divisors. Each $D_i - D_j$ is produced by a search for a weakly holomorphic form with that prescribed Heegner divisor. Two things are then to be checked, and both are checked on the form itself rather than on the search: the denominator n of its principal part, after which Corollary 2 applies — in practice the search returns a rational principal part, its integrality option making the solution integral over an echelonised basis of the pool, which is a weaker thing — and that a form with that principal part exists at all. The second is witnessed by presenting the form as a combination of eta quotients, which is what the search computes and what the existence claim really rests on. The coefficients of that combination are rational, and in practice large — they come out of an echelon form — so the presentation is not short; but it is finite and explicit, and expanding it and recomputing the divisor is cheap compared with the search that produced it. Recording it is therefore what lets a later reader verify the hypothesis without repeating the search.

Remark 6. The count needs r divisors to exist, not that the search found all there are. Exhaustiveness of a search is therefore never part of the argument, which matters because exhaustiveness is the part that is expensive and the part that depends on a chosen window of pole orders.

4. The case that prompted this

Example 7. Let $X = X_0^{46}(31)/W_{46,31}$, of genus three, which has no Atkin–Lehner involution left and so is reached by none of the cover constructions. A search over the degree-two divisors

supported on its rational CM points and its quadratic CM cycles returns exactly six, and they are the six pairwise differences of

$$A = P_{-3} + P_{-403}, \quad B = P_{-24} + P_{-232}, \quad C = P_{-115} + P_{-372}, \quad D = Z(-292),$$

the last a cycle of degree two; the six principal parts are rational with denominators $20, 20, 10, 10, 4, 10$, so Corollary 2 applies to each. The trace formula gives $\#J(\mathbb{F}_p)$ equal to $86, 356, 604, 1882, 3554, 5560$ at $p = 3, 5, 7, 11, 13, 17$, whose greatest common divisor is 2, so $t = 2$. Four effective divisors against $t = 2$ satisfies $r > t$ with room to spare, and Theorem 3 gives that X is hyperelliptic, unconditionally.

Remark 8. Which of the six differences is principal is not determined by the count. For the equation one takes each candidate in turn as the degree-two coordinate, builds the model it gives, and compares the number of points of that model over $\mathbb{F}_p$ with the trace formula, which again needs no prior model and which a wrong candidate fails.

Remark 9. The six divisors have pole orders at infinity between 556 and 1760, and three of them use the quadratic cycle. Neither is reachable by a search restricted to divisors with a single double pole, which is how such searches were previously posed; that restriction returns nothing at all here. It is the availability of several divisors, not of one deeper one, that removes the assumption.

References

[1] R. E. Borcherds, *The Gross–Kohnen–Zagier theorem in higher dimensions*, Duke Math. J. **97** (1999), 219–233.
[2] R. E. Borcherds, *Correction to: "The Gross–Kohnen–Zagier theorem in higher dimensions"*, Duke Math. J. **105** (2000), 183–184.
[3] J.-W. Guo and Y. Yang, *Equations of hyperelliptic Shimura curves*, Compositio Math. **153** (2017), 1–40.
[4] N. M. Katz, *Galois properties of torsion points on abelian varieties*, Invent. Math. **62** (1981), 481–502.